

ROCHES POINT, Ireland

WMO: 039520

Lat: 51.800N

Lon: 8.250W

Elev: 41

StdP: 100.83

Time Zone: 1.00 (IRL)

Period: 01-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.1	1.1	-3.7	2.8	1.4	-2.4	3.1	2.6	20.0	8.8	18.1	8.5	4.7	350	0.740

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.0	20.1	16.7	19.1	16.0	18.2	15.5	17.6	19.2	16.9	18.1	16.3	17.5	5.3	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	12.1	18.2	16.3	11.7	17.5	15.7	11.2	16.9	49.5	19.3	47.4	18.2	45.8	17.5	20.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.3	14.3	12.6	DB	-0.9	22.9	1.6	1.3	-2.1	23.8	-3.0	24.6	-4.0	25.4	-5.1	26.3	(4)
(5)			WB	-1.8	19.3	1.6	1.1	-2.9	20.0	-3.8	20.7	-4.7	21.3	-5.8	22.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	7.0	6.8	7.5	9.3	11.2	13.9	15.3	15.3	14.2	11.9	9.1	7.3	(6)	
	DBStd	3.78	2.40	2.23	2.08	1.81	1.64	1.71	1.77	1.36	1.68	2.12	2.39	2.61	(7)	
	HDD10.0	447	96	92	79	34	7	0	0	0	0	8	43	88	(8)	
	HDD18.3	2771	352	324	335	272	221	134	96	93	125	199	277	343	(9)	
	CDD10.0	719	2	1	2	12	44	116	164	166	125	68	16	3	(10)	
	CDD18.3	2	0	0	0	0	0	0	1	1	0	0	0	0	(11)	
	CDH23.3	1	0	0	0	0	0	0	1	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	6.2	7.2	6.7	6.4	6.0	6.3	5.5	5.4	5.7	5.6	6.4	6.8	7.1	(14)	
(15) Precipitation	PrecAvg	945	104	87	77	55	63	54	56	81	80	97	91	100	(15)	
	PrecMax	1231	220	207	161	179	143	141	123	238	176	206	175	187	(16)	
	PrecMin	685	26	5	3	14	13	16	4	19	5	24	39	16	(17)	
	PrecStd	126	49	50	43	35	36	30	32	45	46	52	42	46	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.4	12.4	13.5	15.9	18.1	20.2	22.4	21.2	19.4	17.0	14.6	12.6	(19)	
		MCWB	11.3	11.1	10.4	11.4	14.3	16.0	18.2	17.7	16.4	14.7	13.5	11.9	(20)	
	2%	DB	11.6	11.4	12.1	13.9	16.2	18.8	20.5	19.6	18.4	16.1	13.7	11.8	(21)	
		MCWB	10.7	10.3	9.8	10.8	13.0	15.3	17.1	16.4	15.8	14.6	12.7	11.0	(22)	
	5%	DB	11.0	10.6	11.1	12.8	14.9	17.7	19.4	18.6	17.5	15.4	13.1	11.4	(23)	
		MCWB	10.2	9.6	9.2	10.6	12.1	14.7	16.4	15.7	15.4	14.2	12.2	10.6	(24)	
	10%	DB	10.5	10.0	10.4	12.0	13.9	16.8	18.3	17.9	16.8	14.9	12.5	10.8	(25)	
		MCWB	9.7	9.0	9.0	10.2	11.5	14.2	15.7	15.2	15.1	13.8	11.6	10.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	11.6	11.3	12.7	15.3	16.9	18.7	18.6	17.5	15.8	14.0	12.2	(27)	
		MCDB	12.0	12.0	12.5	14.0	17.1	18.9	21.0	20.4	18.4	16.1	14.4	12.4	(28)	
	2%	WB	11.0	10.6	10.5	11.9	13.7	16.1	17.7	17.5	16.7	15.2	13.1	11.4	(29)	
		MCDB	11.4	11.1	11.2	12.9	15.3	18.0	19.6	18.7	17.5	15.6	13.4	11.7	(30)	
	5%	WB	10.5	9.9	9.9	11.2	12.8	15.4	17.0	16.7	16.2	14.6	12.5	10.8	(31)	
		MCDB	10.8	10.4	10.5	12.2	14.1	17.0	18.7	17.7	17.0	15.1	12.9	11.2	(32)	
	10%	WB	9.9	9.2	9.4	10.6	12.2	14.8	16.2	16.1	15.6	14.1	11.8	10.3	(33)	
		MCDB	10.3	9.8	10.2	11.6	13.3	16.3	17.8	17.1	16.5	14.7	12.4	10.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.0	4.0	4.6	5.0	5.0	5.2	5.0	5.0	4.8	4.2	4.1	3.6	(35)	
		MCDBR	3.9	4.0	5.0	5.6	6.3	6.3	6.2	6.0	5.2	4.2	4.0	3.7	(36)	
	5% WB	MCWBR	4.2	3.7	4.0	3.9	4.1	4.0	3.8	3.8	3.8	3.5	3.9	4.0	(37)	
		MCWBR	3.9	3.6	4.3	4.9	5.0	5.5	4.8	4.5	4.5	3.6	3.7	3.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.338	0.348	0.372	0.377	0.388	0.372	0.378	0.360	0.357	0.349	0.331	0.329	(40)	
		taud	2.293	2.352	2.307	2.333	2.335	2.411	2.399	2.462	2.465	2.441	2.362	2.263	(41)	
	Ebn at Noon		647	765	816	860	865	882	869	869	829	756	656	583	(42)	
		Edh at Noon	67	82	104	113	118	111	110	99	88	75	62	59	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.78	1.48	2.62	4.20	5.12	5.57	5.06	4.30	3.13	1.79	0.97	0.59	(44)	
	RadStd		0.07	0.12	0.26	0.31	0.31	0.45	0.41	0.25	0.23	0.16	0.07	0.07	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page